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Inadequate Physical Activity and Sedentary Behavior Are Independent Predictors of Nonalcoholic Fatty Liver Disease

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Abbreviations: BMI, body mass index; CI, confidence interval; HSI, hepatic steatosis index; NAFLD, nonalcoholic fatty liver disease; NHANES, National Health and Nutrition Examination Survey; OR, odds ratio; PA, physical activity; US, United States; USFLI, US fatty liver index.

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Abstract

In general, physical activity (PA) and nonalcoholic fatty liver disease (NAFLD) have an inverse association. However, studies assessing the impact of the widely accepted *Physical Activity Guidelines for Americans (PA Guidelines)* on NAFLD are lacking. We performed a serial, cross-sectional analysis among adults by using the 2007-2016 United States National Health and Nutrition Examination Survey. NAFLD and advanced fibrosis were defined by using various noninvasive panels. A PA questionnaire assessed the leisure-time PA, occupation-related PA, transportation-related PA, and total sitting time as sedentary behavior. PA was categorized according to the *PA Guidelines*. Of the 24,588 individuals (mean age 47.4 years; 47.9% males), leisure-time PA (≥ 150 minutes/week) demonstrated 40% lower odds of NAFLD, whereas transportation-related PA was associated with 33% risk reduction in NAFLD. Analysis of total PA and sitting times simultaneously showed a dose-response association between sitting time and NAFLD (P for trend < 0.001). Compliance with the *PA Guidelines* was lower in NAFLD versus non-NAFLD. The trends in compliance with the *PA Guidelines* for any type of PA remained stable in NAFLD except for a downtrend in transportation-related PA. In contrast, an improvement in compliance with the *PA Guidelines* for leisure-time was noted in the non-NAFLD cohort. Although PA demonstrated 10% stronger association with risk reduction of NAFLD in women, women showed a lower tendency of meeting the PA guidelines. Trends in total sitting time increased significantly regardless of NAFLD status. **Conclusion:** Sedentary behavior emerged as an independent predictor of NAFLD. Overall compliance with the *PA Guidelines* was lower in the NAFLD cohort with sex- and ethnicity-based differences. Implementation of these observations in clinical practice may improve our understanding as well as clinical outcomes.

Nonalcoholic fatty liver disease (NAFLD) is the most common chronic liver disease, with a prevalence approaching 30% among adults in the United States.⁽¹⁾ NAFLD-related advanced fibrosis is on the rise in the United States,⁽²⁾ with future projections and forecast of an increase in all-cause and cause-specific mortality in the next decade.^(3, 4) Currently, there are no approved pharmacologic treatments for the management of NAFLD and NAFLD-related advanced fibrosis. Therefore, lifestyle modifications consisting of physical activity (PA), dietary changes, and weight loss are the cornerstone of optimizing the management of patients with NAFLD. There are proven beneficial effects of PA in retarding the progression of NAFLD in the presence or absence of one or more components of metabolic syndrome.⁽⁵⁾ Furthermore, increased PA prevents and/or retards NAFLD-related and comorbidity-associated disease progression independent of weight loss.^(6, 7) Cross-sectional studies have demonstrated an inverse relationship between the prevalence of NAFLD and PA after adjusting for other confounders.^(8, 9) However, most of the studies demonstrating a beneficial effect have focused on leisure-time PA, whereas types, intensity, and duration of PA for optimal therapeutic effects in the management of NAFLD have remained undefined.^(10, 11) Alongside PA, sedentary behavior was associated with increased risk of type 2 diabetes and cardiovascular disease.⁽¹²⁾ Therefore, targeting a reversal of sedentary behavior may complement PA among individuals with NAFLD⁽¹³⁾ because there are no guidelines for sedentary behavior in these individuals.^(10, 11) In terms of ethnological background, several studies reported disparate race/ethnicity-based influence of PA on the prevalence of type 2 diabetes.^(14, 15) Recent emerging data have suggested that NAFLD is a sexually dimorphic disease with sex-based differences impacting various physiological functions and acting as effect modifiers.^(16, 17) In addition, there may be significant behavior-related differences in PA between men and women based on rewards and/or motivation associated with PA and sex hormones.⁽¹⁸⁾ However, clinical evidence to support the contrasting association between PA and sedentary behavior in the setting of NAFLD is lacking.

The 2018 *Physical Activity Guidelines for Americans (PA Guidelines)*⁽¹⁹⁾ recommends that adults should be physically active with moderate intensity for at least 150 to 300 minutes/week or with vigorous intensity for 75 to 150 minutes/week or an equivalent combination of moderate- and vigorous-intensity PA for substantial health benefit.⁽¹⁹⁾ This recommendation emphasizes that the reversal of sedentary behavior will benefit nearly everyone.⁽¹⁹⁾ However, current trends in the

proportions of PA and sedentary behavior among individuals with or without NAFLD have not been established in the United States. In this study, we used nationally representative data with and without NAFLD to estimate recent secular trends in PA and sedentary behavior that were in compliance with the 2018 *PA Guidelines* from 2007 to 2016. Furthermore, we evaluated the association of various types and amounts of PA and sedentary behavior with increased risk for NAFLD and NAFLD-related advanced fibrosis and the interaction of sex and PA with NAFLD. Finally, we investigated the compliance with the 2018 *PA Guidelines* across race/ethnicity and sex to assess for disparities.

Methods

Subjects and Study Design

In our study, we analyzed data from the recent five 2-year waves of the consecutive National Health and Nutrition Examination Survey (NHANES) data from 2007-2008 through 2015-2016, which employ a stratified, multistaged, and clustered probability sampling design to obtain a nationally representative sample of the noninstitutionalized civilians in the United States.⁽²⁰⁾ Of the 29,208 adults (≥ 20 years in age) in the NHANES 2007-2016 data, 96.4% ($n = 28,154$) were examined by laboratory tests at a mobile examination center. Among these, we excluded 3,566 individuals with the hepatitis B virus (determined by the presence of the hepatitis B surface antigen), the hepatitis C virus (determined by the presence of hepatitis C antibodies), significant alcohol consumption (>30 g/day in men and >20 g/day in women), and individuals who have taken medications such as amiodarone, corticosteroids, methotrexate, tamoxifen, and valproate with a known association with fatty liver for more than 6 months; pregnant women; and individuals for whom questionnaires about the PA and data on serum aminotransferase and body mass index (BMI) were not available. Thus, the first study sample using the hepatic steatosis index (HSI) consisted of 24,588 individuals (Supporting Fig. S1). The second study sample using the United States fatty liver index (USFLI) consisted of 11,487 individuals who underwent laboratory tests after an overnight fast for a minimum of 8 hours. This original survey data set was approved by the Centers for Disease Control and Prevention's institutional review board. Individuals included in the analysis provided informed consent. Because the data used in this study were entirely de-identified, this analysis was exempted by the institutional review board.

Clinical and Laboratory Evaluations

The NHANES data included anthropometric measures; comprehensive questionnaires covering sociodemographic and lifestyle information, including PA; and detailed laboratory tests. We categorized race/ethnicity as non-Hispanic white, non-Hispanic black, Hispanic (Mexican American, other Hispanic), or other. Educational attainment was categorized as no high school graduation versus high school graduation. We dichotomized marital status as being married or cohabitating with a partner versus others. Economic status was gauged by the individuals' family income to poverty ratio and categorized as either ≤ 0.99 (below poverty) or ≥ 1.00 (at or above poverty). We calculated alcohol consumption on the amount and frequency of alcohol use with a self-reported questionnaire, as described.⁽²¹⁾ We defined current smokers as individuals who reported ongoing smoking among those who had smoked at least 100 cigarettes in their lifetime. We diagnosed hypertension as systolic blood pressure ≥ 140 mm Hg or diastolic blood pressure ≥ 90 mm Hg and/or treatment with antihypertensive medication. We defined diabetes mellitus as a fasting plasma glucose level ≥ 126 mg/dL and/or use of a hypoglycemic agent or insulin.

PA and Sedentary Behavior Questionnaire

All individuals answered questionnaires providing details of their daily PA based on the Global Physical Activity Questionnaire. The detailed questionnaire assessed the leisure-time PA, occupation-related PA, and transportation-related PA. Each type of PA included questions to assess the intensity (vigorous versus moderate), frequency (per week), and duration (minutes) in a typical week. Individuals reported details of vigorous and/or moderate intensity of PA regarding occupation-related PA and leisure-time-related PA. Minutes of vigorous PA were doubled and added to minutes of moderate PA for occupation-related PA and leisure-time-related PA, as validated.⁽²²⁾ The total amount of PA was defined as the sum of the leisure-time PA, occupation-related PA, and transportation-related PA. PA was categorized according to the 2018 *PA Guidelines* (adults who engage in ≥ 150 minutes/week of moderate-intensity PA, 75 minutes/week of vigorous-intensity PA, or an equivalent combination).⁽¹⁹⁾ We categorized PA as "physically inactive" ("not meeting *PA Guidelines*") when individuals did not meet the criteria outlined in the 2018 *PA Guidelines* and as "physically active" ("meeting *PA Guidelines*") when individuals met the criteria in the 2018 *PA Guidelines*.⁽¹⁹⁾ To investigate sedentary behavior, we assessed total sitting time by the reported hours per day in a typical week. Total sitting time was determined with the survey item, "How much time do you usually spend sitting or reclining on a

typical day (including time spent sitting at a desk, sitting with friends, traveling in a car or bus or train, reading, playing cards, watching television, or using a computer)?”

Definition of NAFLD and advanced fibrosis

For the definition of NAFLD, we used described noninvasive panels.⁽²⁾ Briefly, we defined NAFLD using the HSI and USFLI without other causes of liver disease (positive hepatitis C antibody, positive hepatitis B surface antigen, and/or significant alcohol consumptions) and use of steatogenic medication. Both indices have been validated externally and used in epidemiologic studies.⁽²³⁻²⁶⁾ We calculated HSI by using the following equation: $HSI = 8 \times (\text{alanine aminotransferase/aspartate aminotransferase ratio}) + \text{BMI} (+2 \text{ for female; } +2 \text{ for diabetes}).^{(23)}$ We used the published cutoff of 36 to define presence of NAFLD.⁽²³⁾ Because the USFLI equation requires fasting insulin and glucose, analyses using USFLI included a subgroup of 11,487 individuals who were examined after an overnight fast for a minimum of 8 hours. We calculated the USFLI using the following equation: $USFLI = (e^{-0.8073 * \text{non-Hispanic black} + 0.3458 * \text{Mexican American} + 0.0093 * \text{age} + 0.6151 * \log_e(\text{gamma glutamyltransferase}) + 0.0249 * \text{waist circumference} + 1.1792 * \log_e(\text{insulin}) + 0.8242 * \log_e(\text{glucose}) - 14.7812}) / (1 + e^{-0.8073 * \text{non-Hispanic black} + 0.3458 * \text{Mexican American} + 0.0093 * \text{age} + 0.6151 * \log_e(\text{gamma glutamyltransferase}) + 0.0249 * \text{waist circumference} + 1.1792 * \log_e(\text{insulin}) + 0.8242 * \log_e(\text{glucose}) - 14.7812}) * 100.^{(26)}$ Noninvasive panels for NAFLD (fatty liver index) and advanced fibrosis are shown in the Supporting Methods.

Statistical Analysis

Due to the complex survey design applied by the NHANES, we used appropriate sample weights, stratification, and clustering to ensure representative population-level data for the entire United States.⁽²⁷⁾ We provided frequencies of categorical variables and the means $\pm$ standard errors of the continuous variables. We compared baseline characteristics using the chi-square test for categorical variables or linear regression for continuous variables. The interactions between sex and various types of PA on NAFLD were assessed by examining the significance of the cross-product term of sex and various types of PA. If there were significant interactions, we performed stratified analysis by sex. We performed multivariable logistic regression to identify an independent association between types of PA and total sitting time and NAFLD after consideration of other potential demographic and clinical confounders. To test for trends across cycles (every 2 years), we regressed cycles as an ordinal variable in the outcome of meeting the PA Guidelines and sedentary behavior based on the presence or absence of NAFLD. This test for a

linear trend provides complete correction for the survey design and probability weights of the NHANES data, which is not available in many other trend tests.⁽²⁸⁾ We performed all analyses with STATA 15.1 (StataCorp, College Station, TX, USA) using Taylor series linearization.

Results

A total of 24,588 individuals were eligible and included in the analysis (mean age 47.4 years; 47.9% males). Table 1 shows the baseline characteristics of individuals based on eligibility criteria for PA outlined in the 2018 *PA Guidelines*. There were noticeable differences in the clinical characteristics between the two groups. Compared to physically inactive individuals, physically active individuals (meeting the *PA Guidelines*) were more likely to be younger, male, and educated and less likely to be diabetic, hypertensive, and below poverty. Physically active individuals also had a lower BMI and waist circumference and lower levels of triglycerides and fasting glucose compared to physically inactive individuals.

Results of logistic regression analyses can be seen in Table 2. Leisure-time PA that met the *PA Guidelines* (≥ 150 minutes/week) was inversely associated with the prevalence of NAFLD using HSI as the NAFLD prediction model in the age- and sex-adjusted model (odds ratio [OR], 0.58; 95% confidence interval [CI], 0.53-0.64). After adjusting for multiple confounders, meeting the criteria for leisure-time PA was associated with 40% lower odds of NAFLD. Transportation-related PA that met the guideline (≥ 150 minutes/week) was associated with the lower risk for NAFLD in the age- and sex-adjusted model (OR, 0.66; 95% CI, 0.59-0.73) and the multivariate model (OR, 0.67; 95% CI, 0.60-0.74). The addition of diabetes mellitus and waist circumference to the model reduced the ORs for leisure-time PA and transportation-related PA but did not impact the significance. When we evaluated for C-reactive protein as a mechanistic pathway using the subset of NHANES 2007-2010 data (Supporting Table S1), ORs for leisure-time PA and transportation-related PA were slightly attenuated but remained significant. These results suggest that the association between leisure-time PA or transportation-related PA and NAFLD might be mediated in part by diabetes, inflammation, and abdominal obesity. However, there was no association between occupation-related PA and NAFLD in age- and sex-adjusted models and multivariate models. When we tested the interaction between sex and various types of PA on NAFLD, there were significant interactions between total or leisure-time PA and sex (test of interaction $P < 0.05$) but no significant interactions between occupation-related PA or

transportation-related PA and sex (test of interaction $P > 0.1$) for the risk of NAFLD. Similar and significant associations between total PA, leisure-time PA, and transportation-related PA and NAFLD were, and these associations were more pronounced in women (Table 2 and Supporting Table S2). PA demonstrated an approximately 10% stronger association with a lower risk of NAFLD in women than in men. To investigate whether there is a dose-response relationship between various types of PA and NAFLD and to further assess benefit for PA beyond or below the guidelines, we categorized multiple types of PA into four groups: 0, <150, 150-299, and ≥ 300 minutes/week. Protective effects on NAFLD were observed across different total PA categories (Table 3). When adjusted for all study covariates, individuals performing <150 minutes/week of total PA had a 22% lower risk of NAFLD (OR, 0.78; 95% CI, 0.70-0.86) than physically inactive individuals. Those who reported one or two times (150-299 minutes/week) or more than two times (≥ 300 minutes/week) the recommended level of total PA had 29% (OR, 0.71; 95% CI, 0.62-0.82) and 41% (OR, 0.59; 95% CI, 0.49-0.54) lower risk of NAFLD, respectively. Leisure-time PA was inversely associated with NAFLD in a dose-dependent fashion (P for trend < 0.001) in a multivariate logistic model. The association between transportation-related PA and NAFLD showed a similar tendency (P for trend < 0.001). However, there was no significant relationship between occupation-related PA and NAFLD. When we performed sensitivity analyses using USFLI (Supporting Tables S3-S4), similar and significant associations between total PA, leisure-time PA, and transportation-related PA and NAFLD were observed. When fatty liver index was utilized (Supporting Tables S5 and S6), we observed similar and significant associations between leisure-time PA or transportation-related PA and NAFLD. In addition, sensitivity analyses were performed to exclude individuals who were prescribed anti-steatogenic medications^(29, 30) ($n = 18,261$); and the results were identical to our original overall analyses. In the multivariate-adjusted logistic regression analysis, types of PA demonstrated a statistically significant inverse association with advanced fibrosis (OR, 0.62; 95% CI, 0.45-0.85 [leisure-time PA]; OR, 0.69; 95% CI, 0.55-0.86 [occupation-related PA]; and OR, 0.53; 95% CI, 0.33-0.87 [transportation-related PA]) among individuals with NAFLD (Table 4). Sensitivity analyses were largely identical when we employed the different definitions of NAFLD by using USFLI (Supporting Table S7). Increased sitting time was dose-dependently associated with increased risk for NAFLD in the age- and sex-adjusted model and multivariate model (P for trend < 0.001) (Table 5). When we considered total

PA and sitting time simultaneously, the dose-response association between sitting time and NAFLD was attenuated but maintained (P for trend < 0.001). The addition of waist circumference to the model did reduce the ORs for sitting time. These observations suggest that the association between sitting time and NAFLD might be mediated by abdominal obesity. There were no significant interactions between sitting time and sex for NAFLD (test of interaction $P > 0.2$). In terms of advanced fibrosis, prolonged sitting time was associated with advanced fibrosis among individuals with NAFLD in a dose-dependent manner. After adjustment for total PA, this association remained significant (P for trend = 0.002; Supporting Table S8).

Trends in Various Types of PA and Sedentary Behavior

When we divided the cohort into individuals with and without NAFLD, the prevalence of meeting the *PA Guidelines* for total PA (≥ 150 minutes/week) was markedly higher in individuals without NAFLD than those with NAFLD, which was mostly attributable to the higher prevalence of leisure-time PA in individuals without NAFLD (Table 6). In the total population, the prevalence of meeting the *PA Guidelines* for total PA, leisure-time PA, and occupation-related PA (≥ 150 minutes/week) did not change over time (Fig. 1), whereas transportation-related PA decreased during the study. Among individuals without NAFLD, there was a similar but more pronounced increase in meeting the criteria for leisure-time PA outlined in the 2018 *PA Guidelines* during the study. However, the likelihood of meeting the criteria for leisure-time PA and occupation-related PA remained stable without any significant changes among individuals with NAFLD, whereas the probability of meeting the criteria for transportation-related PA decreased during the study ($P = 0.004$). When we performed sensitivity analyses using USFLI (Supporting Fig. S2), similar trends were observed; however, these trends did not reach significance. From 2007 through 2016, total sitting time increased from 5.5 to 6.5 hours/day in the total population (Supporting Table S9). Prolonged sitting time was observed in individuals with NAFLD compared to those without NAFLD, whereas total sitting time increased regardless of NAFLD status during the study. NAFLD without advanced fibrosis demonstrated a higher probability of meeting the criteria of the *PA Guidelines* for all types of PA than NAFLD with advanced fibrosis (Supporting Fig. S3).

Trends in Types of PA and Sedentary Behavior Across Race/Ethnicity

The prevalence of meeting the PA criteria outlined in the 2018 *PA Guidelines* was stratified by race/ethnicity (Fig. 2A and Supporting Fig. S4A). We noted disparities with significant differences.

When the analysis was stratified by NAFLD status, there was a significant difference in meeting the criteria for leisure-time PA compared to occupation-related PA, and these trends were maintained regardless of NAFLD. There were also significant sex-based differences (Fig. 2B and Supporting Fig. S4B). Across the full spectrum of PA, the prevalence of meeting the *PA Guidelines* was higher in men compared to women. Total sitting time was similar in non-Hispanic whites and non-Hispanic blacks, followed distantly by Hispanics in the third place, regardless of NAFLD status (Supporting Table S10).

Discussion

In this population-based study with a nationally representative sample of US adults, the prevalence of meeting the criteria for PA outlined in the 2018 *PA Guidelines* was lower in individuals with NAFLD than in those without NAFLD. Although the likelihood of meeting the criteria in the *PA Guidelines* for leisure-time increased in individuals without NAFLD from 2007 through 2016, the trends in meeting the criteria based on the *PA Guidelines* for any types of PA remained stable among those with NAFLD, with downtrends in transportation-related PA. In terms of sedentary behavior, the trends increased significantly regardless of NAFLD status during the study time.

Importantly, leisure-time PA and transportation-related PA have a significant dose-dependent protective effect on NAFLD independent of known shared risk factors. Additionally, we noted significant benefits of PA recommendations on NAFLD-related advanced fibrosis. Several meta-analyses have demonstrated a beneficial effect of PA per se on NAFLD.^(31, 32) A significant pooled effect was observed, highlighting the positive influence of PA therapy versus control in improving hepatic fat content without concurrent diet intervention.⁽³¹⁾ Another recent meta-analysis reported quantitative improvement in intrahepatic triglyceride content by using magnetic resonance imaging.⁽³²⁾ In addition, a substantial benefit of PA on NAFLD has been demonstrated in the absence of weight change.⁽³³⁾

A population-based study in Australia studying the trends from 2002 to 2012 showed that sufficient leisure-time PA was slowly increasing, but a substantial proportion of the population was still at increased risk of adverse health outcomes due to inadequate PA.⁽³⁴⁾ To the best of our knowledge, this is the first study to investigate the recent trends (2007-2016) in the prevalence of meeting PA of US adults and their association to NAFLD based on recommendations outlined

in the 2018 *PA Guidelines*. Currently, PA is considered a cornerstone of lifestyle modification in the management of NAFLD. The trends in the prevalence of meeting the *PA Guidelines* for any types of PA remained stable except for the decline in transportation-related PA among individuals with NAFLD. In addition, the estimated total sitting time was longer among individuals with NAFLD than those without NAFLD during the study.

The 2018 *PA Guidelines* recommended a minimum of 75 vigorous-intensity or 150 moderate-intensity PA minutes per week for substantial health benefit and suggested additional benefits by doing more than double this amount.⁽¹⁹⁾ We observed significant benefits for recommended leisure-time PA and transportation-related PA and additional benefit at more than two times for the recommended PA minimum. Our study showed that a low dose of PA, which was less than the recommended minimum by the *PA Guidelines*, was associated with reduced risk for NAFLD compared to physical inactivity. Data on the intensity and duration of PA were obtained from diverse and heterogeneous populations; hence, it is difficult to make a conclusive recommendation.⁽³⁵⁾ Our study suggests that clinicians may encourage inactive adults to perform PA according to current PA guidelines for the substantial benefit of improving NAFLD in the United States. The 2018 *PA Guidelines* addressed the risks of prolonged sedentary behavior in adults for the first time but does not recommend an essential quantitative guideline for sitting time.⁽¹⁹⁾ Given the low adoption of the 2018 *PA Guidelines* in the United States, increasing PA and decreasing sedentary time are both likely to provide benefits.⁽¹⁹⁾ Our study trends from 2007 to 2016 clearly demonstrate that US adults spend a considerable amount of time during their day engaged in sitting and other sedentary behaviors. In this study, sitting more than 8 hours/day was associated with increased risk for NAFLD independent of increasing PA. Thus, we can assume that sedentary behavior and PA have independent effects on NAFLD. Therefore, reducing the sitting time in a day may be a modifiable option as one lifestyle modification.

Across all types of PA, there was a consistent pattern in which NAFLD prevalence was lower among individuals who met the criteria for the *PA Guidelines* than those who did not. PA demonstrated an approximately 10% stronger association with a lower risk of NAFLD in women than in men, whereas the prevalence of meeting the *PA Guidelines* was lower in women.

Race/ethnicity-based disparities were observed and negatively impacted minorities. When comparing differences in the prevalence of meeting the *PA Guidelines* by NAFLD, the reductions

were higher in non-Hispanic whites across all domains except occupation-related PA, in which the prevalence reduction was somewhat similar among non-Hispanic blacks and Hispanics. Our results demonstrate race/ethnicity-based differential benefits of PA on the prevalence of NAFLD, similar to studies focusing on diabetes.^(14, 15) The protective effects of PA on NAFLD were identical across all domains of PA except occupation-related PA. Our findings highlight the importance of examining the various types of PA to understand the potential etiologic role of PA on NAFLD, especially given the sex- and race/ethnicity-related disparities observed in different types of PA. The strengths of this study are the use of validated PA and sedentary behavior questionnaires collected by trained personnel with a systematic protocol, the wealth of metabolic variables, and a large cohort size which was representative of the US general population. Therefore, our findings are generalizable to the US population regardless of race/ethnicity. However, we did encounter some limitations in our study. First, we used three noninvasive panels to define NAFLD in the absence of co-existing liver disease and history of exposure to steatogenic medication, which may misclassify and under/overestimate the true prevalence of NAFLD. However, by employing three noninvasive panels, we hope that the likelihood of misclassification was minimized. These panels have been validated as reliable noninvasive panels for NAFLD in the US population and other populations and are independently associated with all-cause and liver-related mortality.^(23-26, 36-40) Second, the temporal causality of the observed association could not be established due to the cross-sectional study design. Given that the bidirectional association of PA with NAFLD and NAFLD-related advanced fibrosis is unknown, lower PA can result from NAFLD and advanced fibrosis due to reverse causality. Third, we retrieved the PA status from a self-reported questionnaire rather than objectively measured PA, although this PA questionnaire was based on the well-validated Global Physical Activity Questionnaire. Finally, the NHANES data do not provide the follow-up PA questionnaire and laboratory data. Therefore, we are unable to assess longitudinal changes in NAFLD or PA status in this study.

In summary, our study suggests that increasing leisure-time PA and transportation-related PA are protective factors for NAFLD, whereas the trends in the prevalence of meeting *PA Guidelines* remained stable in NAFLD from 2007 to 2016. The trends in sedentary behavior, which is an independent risk factor for NAFLD, increased significantly in this US population during the recent decade. These data suggest that increasing PA and lowering sedentary behavior need to be

explored as a potential strategy for treatment and/or prevention of NAFLD. Currently, there are no approved medical therapies for the management of NAFLD, and lifestyle modifications remain the first-line treatment of patients with NAFLD. A multidisciplinary approach is warranted to develop consensus among providers and optimize the impact of the 2018 *PA Guidelines* and its future versions.

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Figure Legends

FIG. 1. The trend in prevalence of meeting the physical activity guideline by year (2007-2016) in the United States. We categorized physical activity as physically active individuals who met the guideline according to the 2018 *Physical Activity Guidelines for Americans* (adults engage in ≥ 150 minutes/week of moderate-intensity activity, 75 minutes/week of vigorous-intensity activity, or an equivalent combination). NAFLD was defined using the hepatic steatosis index. Abbreviations: LTPA, leisure-time physical activity; NAFLD, nonalcoholic fatty liver disease; OPA, occupation-related physical activity; PA, physical activity; TPA, transportation-related physical activity.

FIG. 2. Prevalence of meeting the physical activity guideline by race/ethnicity and sex in the United States. (A) Race/ethnicity. (B) Sex. We categorized physical activity as physically active individuals who met the guideline according to the 2018 *Physical Activity Guidelines for Americans* (adults engage in ≥ 150 minutes/week of moderate-intensity activity, 75 minutes/week of vigorous-intensity activity, or an equivalent combination). NAFLD was defined using the hepatic steatosis index. Abbreviations: NAFLD, nonalcoholic fatty liver disease; PA, physical activity.

TABLE 1. Characteristics of Study Participants Based on Total Physical Activity Status (n = 24,588)

	Physically Inactive (n = 10,026)	Physically Active (n = 14,562)	P Value
Age (years)	52.2 ± 0.3	44.7 ± 0.3	<0.001
Sex (male)	37.7 (36.5-39.0)	53.7 (52.9-54.5)	<0.001
Body mass index (kg/m ²)	30.1 ± 0.1	28.3 ± 0.1	<0.001
Waist circumference (cm)	100.0 ± 0.3	96.5 ± 0.3	<0.001
Hypertension (%)	41.0 (39.7-42.3)	25.2 (23.9-26.6)	<0.001
Diabetes (%)	16.4 (15.5-17.3)	7.8 (7.2-8.4)	<0.001
Ethnicity (%)			<0.001
Non-Hispanic white	64.8 (61.0-68.4)	68.5 (65.0-71.7)	
Non-Hispanic black	11.7 (10.0-13.7)	9.7 (8.3-11.3)	
Hispanic	15.4 (12.8-18.3)	13.8 (11.8-16.1)	
Other	8.1 (7.0-9.4)	7.3 (6.4-8.2)	
Smoking (%)			0.318
Never	55.8 (54.2-57.4)	56.8 (55.2-58.5)	
Current smoker	19.0 (17.6-20.5)	19.2 (18.0-20.4)	
Ex-smoker	25.2 (24.0-26.5)	24.0 (22.8-25.2)	
High school graduation (%)	78.4 (76.6-80.2)	86.1 (84.7-87.5)	<0.001
Marital status (%)	63.1 (61.6-64.6)	64.6 (62.9-66.2)	0.105
Poverty (%)	16.4 (14.8-18.0)	13.7 (12.5-15.0)	<0.001
Total cholesterol (mg/dL)	194.6 ± 0.6	194.3 ± 0.6	0.652
Fasting triglycerides (mg/dL)	135.8 ± 2.4	125.6 ± 2.1	0.001
High-density lipoprotein cholesterol (mg/dL)	52.2 ± 0.3	53.6 ± 0.3	<0.001
Fasting glucose (mg/dL)	110.6 ± 0.7	104.1 ± 0.4	<0.001
Hemoglobin A1c (%)	5.8 ± 0.02	5.5 ± 0.01	<0.001
Fasting insulin (pmol/L)	92.5 ± 1.7	73.1 ± 1.4	<0.001
Alanine aminotransferase (IU/L)	24.4 ± 0.2	25.4 ± 0.2	0.001
Aspartate aminotransferase (IU/L)	24.9 ± 0.2	25.5 ± 0.1	0.016
Gamma-glutamyltransferase (IU/L)	27.8 ± 0.5	25.7 ± 0.4	0.016

Data are expressed as the weighted mean $\pm$ standard error or weighted frequency (95% confidence intervals).

We categorized physical activity as physically inactive individuals who did not meet the guideline and physically active individuals who met the guideline according to the 2018 *Physical Activity Guidelines for Americans* (adults engage in ≥ 150 minutes/week of moderate-intensity activity, 75 minutes/week of vigorous-intensity activity, or an equivalent combination).

Fasting triglycerides and fasting glucose/insulin were analyzed from fasting subjects only.

TABLE 2. Multivariate Odds Ratio for NAFLD Based on Meeting the Physical Activity Guideline

	Age/Sex-Adjusted Model		Multivariate Model 1		Multivariate Model 2	
	OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Total physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.65 (0.60-0.71)	<0.001	0.68 (0.62-0.74)	<0.001	0.79 (0.71-0.87)	<0.001
Men						
<150	1		1		1	
≥150	0.73 (0.64-0.82)	<0.001	0.73 (0.65-0.83)	<0.001	0.90 (0.76-1.06)	0.209
Women						
<150	1		1		1	
≥150	0.60 (0.54-0.66)	<0.001	0.64 (0.58-0.72)	<0.001	0.73 (0.63-0.85)	<0.001
Leisure-time physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.58 (0.53-0.64)	<0.001	0.60 (0.55-0.66)	<0.001	0.75 (0.67-0.84)	<0.001
Men						
<150	1		1		1	
≥150	0.64 (0.57-0.72)	<0.001	0.63 (0.55-0.73)	<0.001	0.82 (0.71-0.96)	0.014
Women						
<150	1		1		1	
≥150	0.53 (0.47-0.60)	<0.001	0.58 (0.51-0.66)	<0.001	0.70 (0.59-0.83)	<0.001
Occupation-related physical activity						
(minutes/week)						
<150	1		1		1	
≥150	1.05 (0.97-1.15)	0.231	1.05 (0.96-1.16)	0.270	1.01 (0.92-1.12)	0.778

Men						
<150	1		1		1	
≥150	1.10 (1.00-1.21)	0.045	1.08 (0.97-1.20)	0.170	1.04 (0.92-1.18)	0.533
Women						
<150	1		1		1	
≥150	1.00 (0.89-1.13)	0.967	1.02 (0.90-1.16)	0.754	0.98 (0.84-1.15)	0.823

**Transportation-related physical activity
(minutes/week)**

<150	1		1		1	
≥150	0.66 (0.59-0.73)	<0.001	0.67 (0.60-0.74)	<0.001	0.75 (0.65-0.87)	<0.001
Men						
<150	1		1		1	
≥150	0.66 (0.58-0.75)	<0.001	0.70 (0.61-0.81)	<0.001	0.83 (0.69-1.00)	0.056
Women						
<150	1		1		1	
≥150	0.66 (0.59-0.73)	<0.001	0.64 (0.55-0.75)	<0.001	0.68 (0.55-0.84)	<0.001

The multivariate model 1 was adjusted for age, sex, race/ethnicity, education level, marital status, economic status, smoking status, hypertension, and total cholesterol using appropriate sampling weights.

The multivariate model 2 was adjusted for diabetes and waist circumference in addition to model 1 using appropriate sampling weights.

Abbreviations: CI, confidence interval; NAFLD, nonalcoholic fatty liver disease; OR, odds ratio.

TABLE 3. Multivariate Odds Ratio for NAFLD Based on the Amount of Physical Activity

		Age/Sex-Adjusted Model		Multivariate Model 1		Multivariate Model 2	
		OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Total physical activity							
(minutes/week)							
0	1		<0.001*	1	<0.001*	1	<0.001*
1-149		0.77 (0.70-0.85)	<0.001	0.78 (0.70-0.86)	<0.001	0.91 (0.81-1.02)	0.093
150-299		0.69 (0.61-0.79)	<0.001	0.71 (0.62-0.82)	<0.001	0.87 (0.73-1.03)	0.121
≥300		0.57 (0.52-0.63)	<0.001	0.59 (0.53-0.65)	<0.001	0.74 (0.66-0.82)	<0.001
Leisure-time physical activity							
(minutes/week)							
0	1		<0.001*	1	<0.001*	1	<0.001*
1-149		0.77 (0.70-0.85)	<0.001	0.78 (0.70-0.87)	<0.001	0.88 (0.77-1.01)	0.065
150-299		0.71 (0.64-0.79)	<0.001	0.72 (0.64-0.81)	<0.001	0.88 (0.76-1.02)	0.083
≥300		0.47 (0.43-0.52)	<0.001	0.49 (0.44-0.54)	<0.001	0.65 (0.58-0.74)	<0.001
Occupation-related physical activity							
(minutes/week)							
0	1		0.222*	1	0.226*	1	0.565*
1-149		0.97 (0.86-1.08)	0.547	1.00 (0.89-1.13)	0.983	1.08 (0.90-1.30)	0.397
150-299		0.97 (0.82-1.14)	0.687	0.97 (0.82-1.14)	0.689	0.94 (0.77-1.16)	0.559
≥300		1.06 (0.97-1.17)	0.175	1.07 (0.97-1.18)	0.178	1.04 (0.93-1.16)	0.478
Transportation-related physical activity							
(minutes/week)							
0	1		<0.001*	1	<0.001*	1	0.001*
1-149		0.78 (0.71-0.87)	<0.001	0.81 (0.72-0.90)	<0.001	0.90 (0.77-1.05)	0.197
150-299		0.57 (0.49-0.66)	<0.001	0.59 (0.51-0.70)	<0.001	0.63 (0.53-0.76)	<0.001
≥300		0.68 (0.60-0.78)	<0.001	0.68 (0.59-0.79)	<0.001	0.82 (0.68-1.00)	0.055

The multivariate model 1 was adjusted for age, sex, race/ethnicity, education level, marital status, economic status, smoking status, hypertension, and total cholesterol using appropriate sampling weights.

The multivariate model 2 was adjusted for diabetes and waist circumference in addition to model 1 using appropriate sampling weights.

**P* value for the test of trend of odds.

Abbreviations: CI, confidence interval; NAFLD, nonalcoholic fatty liver disease; OR, odds ratio.

TABLE 4. Multivariate Odds Ratio for Advanced Fibrosis Based on Meeting the Physical Activity Guideline Among Individuals With NAFLD

	Age/Sex-Adjusted Model		Multivariate Model 1		Multivariate Model 2	
	OR (95% CI)	<i>P</i> Value	OR (95% CI)	<i>P</i> Value	OR (95% CI)	<i>P</i> Value
Total physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.62 (0.50-0.76)	<0.001	0.64 (0.51-0.80)	<0.001	0.69 (0.55-0.85)	<0.001
Leisure-time physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.61 (0.45-0.83)	0.002	0.62 (0.45-0.85)	0.004	0.67 (0.48-0.92)	0.016
Occupation-related physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.66 (0.53-0.81)	<0.001	0.69 (0.55-0.86)	0.001	0.71 (0.57-0.88)	0.003
Transportation-related physical activity						
(minutes/week)						
<150	1		1		1	
≥150	0.54 (0.34-0.87)	0.011	0.53 (0.33-0.87)	0.013	0.55 (0.34-0.91)	<0.001

The multivariate model 1 was adjusted for age, sex, race/ethnicity, education level, marital status, economic status, smoking status, hypertension, and total cholesterol using appropriate sampling weights.

The multivariate model 2 was adjusted for diabetes and waist circumference in addition to model 1 using appropriate sampling weights.

Abbreviations: CI, confidence interval; NAFLD, nonalcoholic fatty liver disease; OR, odds ratio.

TABLE 5. Multivariate Odds Ratio for NAFLD Based on the Amount of Sitting Time

	Age/Sex-Adjusted Model		Multivariate Model 1		Multivariate Model 2	
	OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Sitting time						
(hours/day)						
Q1 (<4)	1	<0.001 *	1	<0.001 *	1	<0.001 *
Q2 (≥4 to <6)	1.01 (0.90-1.12)	0.912	1.00 (0.90-1.12)	0.969	0.99 (0.89-1.11)	0.891
Q3 (≥6 to <8)	1.13 (1.02-1.26)	0.025	1.15 (1.02-1.30)	0.024	1.10 (0.98-1.25)	0.106
Q4 (≥8)	1.26 (1.16-1.37)	<0.001	1.32 (1.21-1.43)	<0.001	1.21 (1.11-1.31)	<0.001
Total physical activity						
(minutes/week)						
0					1	<0.001 *
1-149					0.79 (0.71-0.87)	<0.001
150-299					0.73 (0.63-0.84)	<0.001
≥300					0.62 (0.56-0.68)	<0.001

The multivariate model 1 was adjusted for age, sex, race/ethnicity, education level, marital status, economic status, smoking status, hypertension, and total cholesterol using appropriate sampling weights.

The multivariate model 2 was adjusted for total physical activity in addition to model 1 using appropriate sampling weights.

*P value for the test of trend of odds.

Abbreviations: CI, confidence interval; NAFLD, nonalcoholic fatty liver disease; OR, odds ratio; Q, quartile.

TABLE 6. Prevalence of Meeting the Physical Activity Guideline Between Individuals With and Without NAFLD

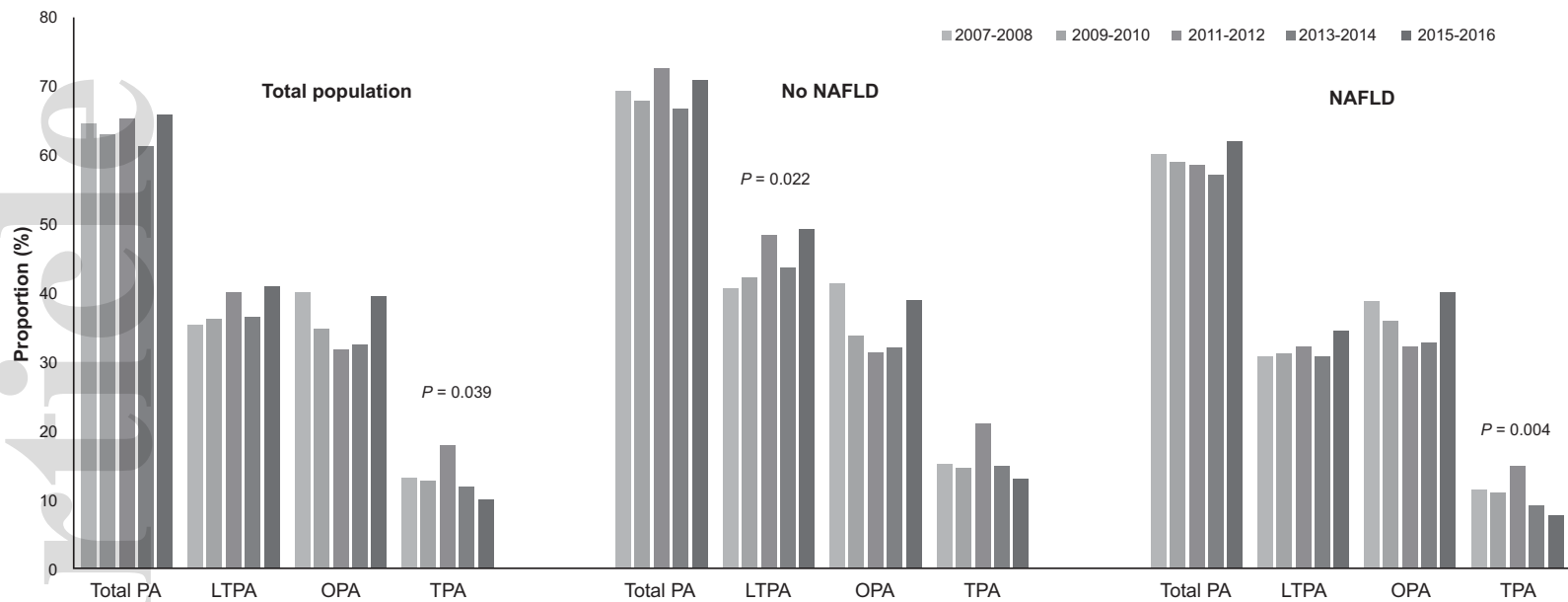
	Total Population	No NAFLD	NAFLD	<i>P</i> Value*
Hepatic steatosis index (n = 24,588)				
Total physical activity	63.9 (62.8-65.1)	69.4 (67.9-70.9)	59.3 (57.9-60.7)	<0.001
Leisure-time physical activity	37.8 (36.2-39.5)	44.9 (42.6-47.2)	31.9 (30.4-33.4)	<0.001
Occupation-related physical activity	35.7 (34.5-37.0)	35.4 (33.8-37.1)	35.9 (34.4-37.5)	0.609
Transportation-related physical activity	13.0 (12.0-14.1)	15.8 (14.4-17.3)	10.7 (9.8-11.7)	<0.001
US fatty liver index (n = 11,487)				
Total physical activity	64.4 (63.0-65.7)	67.9 (66.3-69.4)	57.2 (55.1-59.2)	<0.001
Leisure-time physical activity	36.8 (34.9-38.8)	42.1 (39.8-44.3)	26.1 (24.1-28.3)	<0.001
Occupation-related physical activity	36.3 (34.7-37.9)	36.0 (34.1-38.1)	36.8 (34.6-39.1)	0.611
Transportation-related physical activity	13.6 (12.3-15.0)	14.9 (13.3-16.9)	11.0 (9.5-12.7)	<0.001

Data are expressed as weighted frequency (95% confidence intervals).

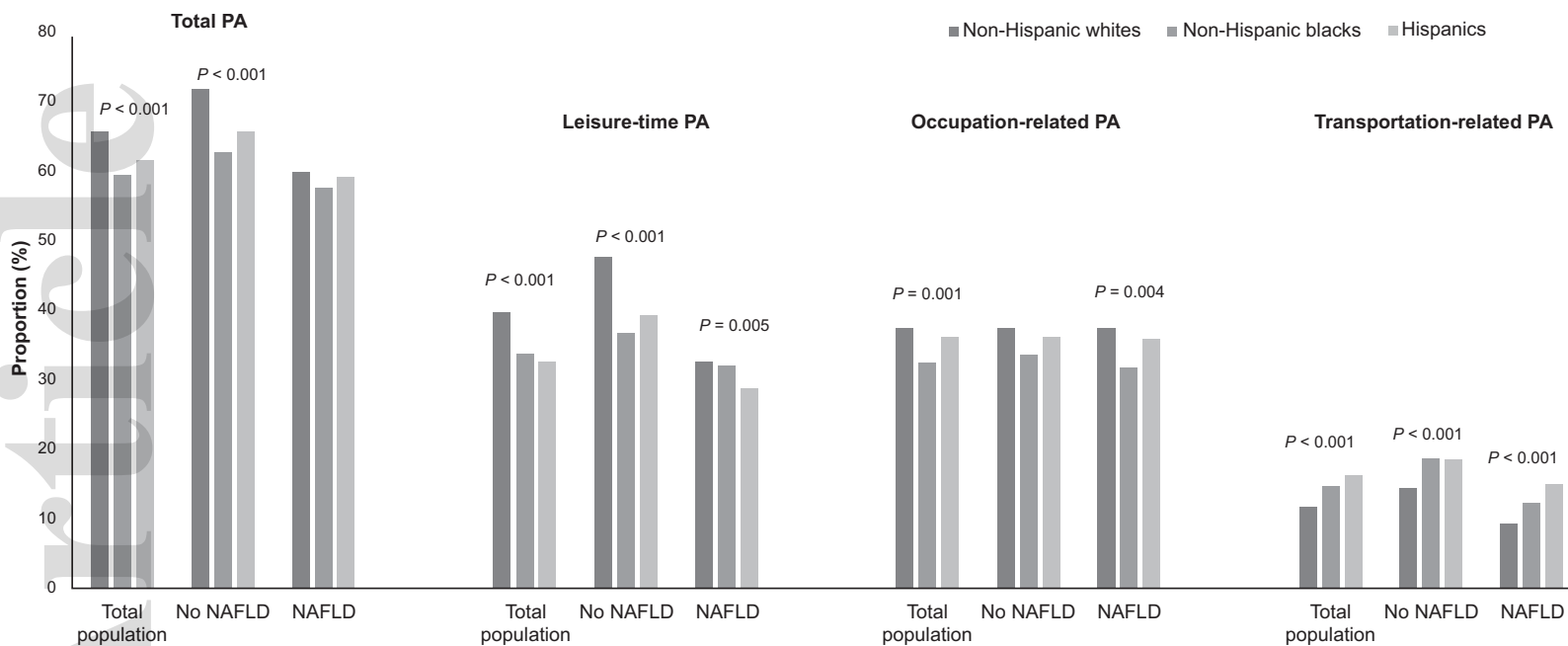
We categorized physical activity as physically active individuals who met the guideline according to the 2018 *Physical Activity Guidelines for Americans* (adults engage in ≥ 150 minutes/week of moderate-intensity activity, 75 minutes/week of vigorous-intensity activity, or an equivalent combination).

**P* value for comparison between individuals with and without NAFLD.

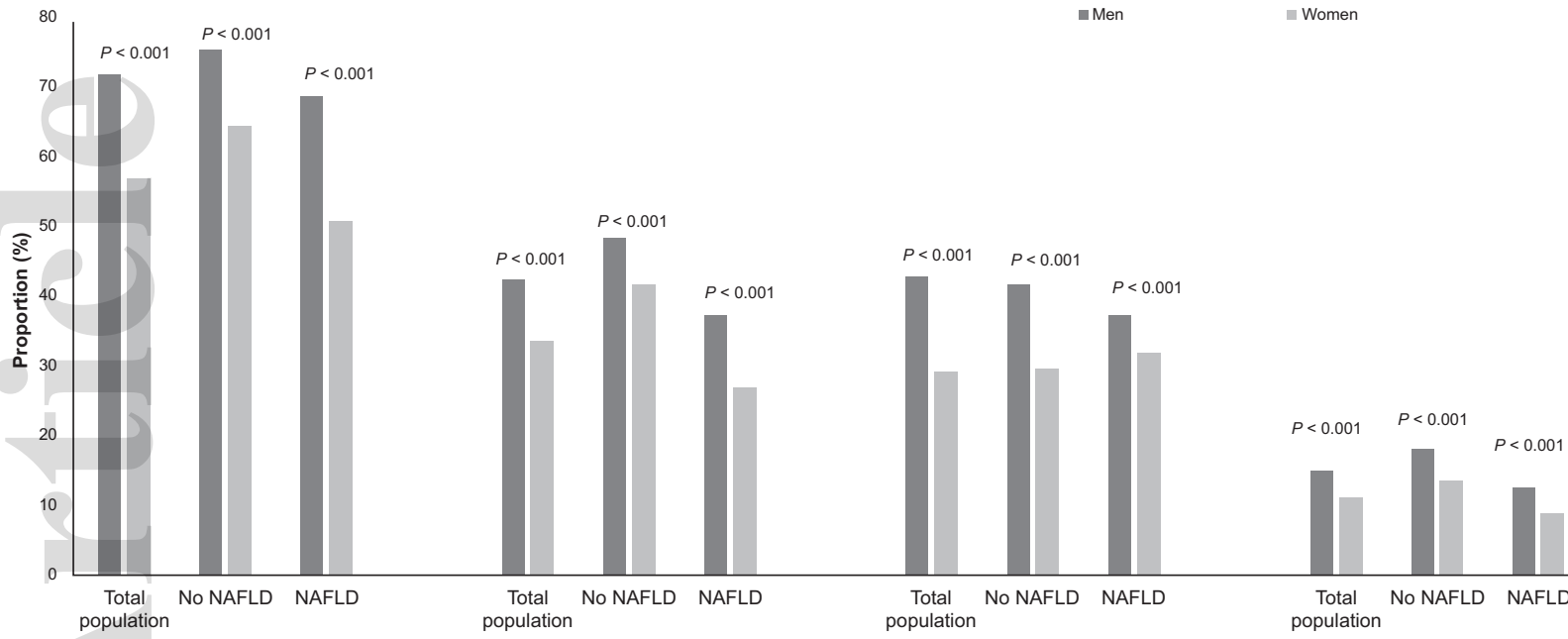
Abbreviations: NAFLD, nonalcoholic fatty liver disease.



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hep_31158_f2a.eps



hep_31158_f2b.eps